

Plant somatic mutations in nature conferring insect and herbicide resistance

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Abstract

Because of the role of the meristem in plant growth and reproduction, somatic mutations in plants have long been suspected of conferring herbivore and pathogen resistance on individual plants and, in the case of trees, individual branches within single plants. A few instances of resistance to phytophagous insects owing to somatic mutations have been reported in the literature. More recently, a striking example has demonstrated how somatic mutations confer resistance to an herbicide on an invasive plant, *Hydrilla verticillata*. The array of new methods for manipulating genomes (e.g., gene-editing) plus existing examples of somatic mutation-associated resistance suggest that such mutations might be useful in silviculture, agriculture, and horticulture. Answering several general questions about somatic mutations in plants would facilitate such applications: Why are so few examples reported? Do other cases exist but go undetected for want of adequate attention or methods? Under what circumstances do somatic mutations enter gametophytes?

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1 SOMATIC MUTATIONS IN PLANTS

Somatic mutations in plants have long interested botanists, particularly from the standpoint of evolution, because, unlike in animals, somatic mutations can be passed on to progeny.^{1,2} Though somatic, an apical meristem mutation can, depending on the layer in which it occurs, enter gametes. In angiosperms, for instance, gametes are derived from the L2 layer in dicot apices, but in monocots the megasporocytes are derived from the L2 layer and the microsporocytes are derived from the L3 layer or from both L2 and L3.¹ However, mutations arising in L1 can, under certain circumstances, pass into the gametes,¹ and cell divisions in the L3 layer can sometimes push L3 cells into the L2 layer.²

Formation of adventitious buds from axillary meristems can also allow somatic mutations that would not normally be found in gametes to become part of the gametophytic generation.³ Herbivore damage in an axillary meristem of *Nicotiana sylvestris* can lead to cells that would not normally have generated gametes nevertheless producing seeds in axillary buds; such a mechanism might exist in other plants.⁴ In gymnosperms, the meristem has two distinct layers, of which the surface (L1) layer is divided into a summit zone at the tip and a surrounding peripheral zone.⁵ At least in conifers, gametes derive only from cells in the L1 summit zone.⁵ It has not been demonstrated that mutations arising elsewhere can enter gymnosperm gametes.

Even if a somatic mutation does not ultimately enter a gamete, in a clonal plant it may be perpetuated indefinitely by vegetative reproduction (e.g., through stolons, tubers, rhizomes, etc.). Many tree species, such as quaking aspen (*Populus tremuloides*) and black locust (*Robinia pseudoacacia*), produce individual trees through cloning.³ Importantly, long-lived individual trees are essentially aerial clones² with individual branches within one tree potentially differing genetically owing to mutations that are subsequently

incorporated as the tree grows. Because branches of the same tree are inextricably linked and can exchange nutrients, even a branch whose reproductive contribution is minimal can persist as long as the tree lives. For instance, a branch system comprising 15% of a chimeric red mangrove (*Rhizophora mangle*) tree produced embryos that were all albino and died after rooting, yet the branch system continued to be maintained.⁶ In some tree species (e.g., variegated maple, *Acer platanoides drummondii*),³ over a period of years branches with certain genotypes are replaced by branches with other genotypes. In horticulture and agriculture, somatic mutations have been intentionally maintained in clones of many plant species because they produce desirable features,³ but the possibility of deliberately using such mutations in trees to produce insect resistance has barely been broached.

Somaclonal variation resulting from somatic mutations in tissue culture has produced commercially exploited varieties in some crop plant species that resist herbicides,⁷ insects,⁸ and plant pathogens.⁸ However, this approach is challenged by ongoing variation in cultured clones,⁸ and commercial use of herbicide-resistant varieties has been limited in some cases by the success of transgenic competitors.⁷ Although the many instances of commercial somaclonal varieties support the view that somatic mutations in nature merit greater study, they do not bear directly on the question of why a relative dearth of them has been reported in the literature.

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2 SOMATIC MUTATION-BASED RESISTANCE TO INSECTS

In the early 1980s, laboratories of Thomas Whitham^{3,9} and Douglas Gill^{10,11} independently explored implications of the fact that trees can be viewed as clones, and, because of somatic mutations, branches may differ with respect to resistance or susceptibility to insect attack. One might not expect within-tree, between-branch variation in trees with strong apical dominance (e.g., most but not all conifers) because they rarely clone.

The main interest of Whitham and Gill along these lines was to try to understand, from an evolutionary standpoint, how long-lived organisms such as trees could survive constant attack from phytophagous insects, with their much shorter generation times and huge populations. The ensuing hypothesis involving somatic mutations came to be known as the “genetic mosaic theory” of plant defense, that trees survived partly because they compensated for their long generation times by the frequent advent of somatic mutations. In support of the genetic mosaic theory, Whitham depicted great variation in attack by the aphid *Pemphigus betae*, persistent over at least 2 years, among branches within individual *Populus angustifolia* trees.⁹ The attack rate correlated strongly with leaf size, and, although the branches were not genotyped, the pattern was highly consistent with somatic mutations differentiating branches.

In a more recent relevant example, attack rates of Christmas beetles (*Anoplognathus* spp.) on yellow box (*Eucalyptus melliodora*) vary greatly both between trees and among branches within individual trees,¹² the latter variation extremely striking (single unattacked branches within otherwise heavily attacked trees) and suggestive of somatic mutation because the attack rate was related to chemicals produced under known genetic control. In a similar mosaic individual of *Eucalyptus sideroxylon* as well as that of *E. melliodora*, Padovan *et al.* found chemical differences between the leaves of the resistant and susceptible branches that matched chemical differences between leaves of resistant and susceptible trees of the same two species.¹³ Studying gene expression in mosaic *E. melliodora* and *E. sideroxylon* trees, Padovan *et al.* found evidence in each species implicating a “master switch” that can cause major phenotypic changes, including in resistance to herbivores.^{14,15}

Many mechanisms other than somatic mutations cause differential rates of attack by phytophagous insects within individual trees.¹⁶ These include chemical changes induced in leaves (both damaged and undamaged) induced by herbivory, chemical and physical differences between old and new leaves, and differences in initial attack owing simply to location of leaves with respect to sunlight and height in the canopy. However, none of the numerous reports except those on *Eucalyptus* and *Populus angustifolia* are consistent with the hypothesis of somatic mutations causing inter-branch differences. In particular, they do not report the same branches heavily attacked year after year.

Thirty years later, the genetic mosaic theory has not fared well in the sense that genetic mosaicism is not now widely seen as a frequent key factor in plant responses to herbivory,¹⁷ but there is little doubt that mosaicism sometimes does characterize trees. For instance, it is common in western red cedar (*Thuja plicata*), a long-lived tree.¹⁸ And it is quite likely that more cases exist but have not been noticed or studied because they do not yield highly visible phenotypic differences, such as albinism or strikingly different leaf color or shape that might be variable in expression,¹⁵ thus they are even less likely to be recognized as patterns.

What is really at issue is the role of somatic mutations in defense against herbivores. In addition to the *Populus* and *Eucalyptus* examples described above, we believe we may have found another case. The hemlock woolly adelgid, *Adelges tsugae*, native to Asia and possibly western North America,¹⁹ has devastated populations in eastern North America of eastern hemlock (*Tsuga canadensis*).²⁰ Studying an eastern hemlock stand in an area of western North Carolina in which most eastern hemlocks are heavily attacked, we have noted that adelgids on some 10-year-old trees attack all but a few branches amidst heavily attacked branches, the same branches for two consecutive years. The relatively adelgid-free branches are not clumped but are well-dispersed – in short, they bear the earmarks of somatic mutation.

As early as 1966, Reginald Painter suggested exploiting between-branch genetic differences to produce resistant trees,²¹ but to our knowledge no sustained research has followed this suggestion. For instance, it seems possible that resistant individuals could be produced through grafting or planting cuttings. If a somatic mutation can enter gametophytic tissue, as in the red mangrove chimera described above,⁶ the seeds could be harvested. That the red mangrove seedlings were inviable rather than resistant to an insect does not mean that a different scenario could not arise involving a similarly produced chimera. A potential long-term problem in wholly vegetatively reproducing branches or individuals is the gradual build-up of other somatic mutations, mostly deleterious.^{22,23}

3 SOMATIC MUTATION-BASED RESISTANCE TO HERBICIDES

In 2004, in Florida, the highly invasive Asian aquatic plant *Hydrilla verticillata* (hydrilla), previously well controlled by the herbicide fluridone, was reported to have become resistant to fluridone in several lakes.²⁴ What is remarkable is that, in Florida, hydrilla is an all-female clone reproducing only vegetatively, so the resistance arose from somatic mutations, in this instance three different mutations of the same gene, and in fact the same codon, in different lakes. Because several other prominent plant invaders (e.g., giant reed [*Arundo donax*], water hyacinth [*Eichhornia crassipes*], phragmites [*Phragmites australis*], Japanese knotweed [*Fallopia japonica* var. *japonica*]) also spread vegetatively, the question arises as to why this event with hydrilla is the first report of such a phenomenon. Further, for each of the three mutations, the mutant form quickly dominated the lakes in which it arose, even when fluridone use was terminated,²⁴ leading to the further question of how this rapid spread occurred.

With respect to why the hydrilla case is the first such report, it is possible that even though much (or in some locations almost all) of the invasions by phragmites,²⁵ water hyacinth,²³ and giant reed²⁶ is by vegetative reproduction, occasional sexual reproduction generates so much more genetic variation than somatic mutations in the vegetatively reproducing individuals that the latter have gone unnoticed. Perhaps the Japanese knotweed invasion of Europe, which is by a female clone that reproduces solely vegetatively,²⁷ is the most analogous to the hydrilla invasion. The fact that ramets of this clone hybridize extensively with sexually reproducing congeneric species²⁷ may partly explain the huge extent of the invasion, but the puzzling fact remains that many large areas are solely invaded by the clone.^{27,28} Somatic mutations have been proposed as a possible explanation for substantial genetic diversity of knotweed in parts of Europe,²⁹ but no somatic mutations have

been reported in three studies that have used AFLP markers,^{28–30} and use of various other methods might reveal genetic variation possibly associated with somatic mutations that have gone undetected.²⁸ In Poland, all recorded Japanese knotweed consists of female clones, but there is more than one clone,³⁰ suggesting either separate invasions or somatic mutations.

To an extent, the hypothesis of occasional sexual reproduction in other vegetatively spreading invasive plants begs the question of why hydrilla in the first place, despite its clonal nature in Florida. In hydrilla the resistance is generated by mutations in the gene *pds* for the enzyme phytoene desaturase that is involved in producing pigments protective to photosynthetic processes.²⁴ Fluridone inhibits phytoene desaturase, and without phytoene desaturase, chloroplast destruction leads to bleaching under high light conditions. Hydrilla can grow in very low light, but plant segments with the mutant *pds* at the surface could survive at high light intensity with fluridone, while supported by unmutated segments lower in the water that would not be faced with the risk of bleaching in low light.²⁴ Hydrilla also can grow very rapidly, thus facilitating the spread of somatic mutations.³¹ Additionally, high light conditions near the surface likely destroy apical meristems, releasing axillary meristems from dormancy,²⁴ a possible source of these mutations. Further, there is no measurable difference in the specific activity of phytoene desaturase isoenzymes produced by the mutants compared to wild-type hydrilla, so the mutant forms may be competitive even in the absence of fluridone, aiding their spread once present.³²

The advent of fluridone-resistant hydrilla in Florida is of great concern, as fluridone has been the main control of a very widespread plant. The resistant genotypes are also resistant to the alternative herbicide norflurazon,³³ also a phytoene desaturase inhibitor, and there is a distinct possibility that the resistant genotypes will spread to other parts of the US South.²⁴ Conversely, it is possible that mechanisms underlying the hydrilla mutations could be used to produce herbicide resistance in crops and other desirable plants, much as transgenic glyphosate-resistant crops are now used.³² This prospect might be realized by gene-editing or use of a transgene.

4 CONCLUDING REMARKS

Somatic mutations, mosaics, and chimeras have long exerted a fascination if only because of some of the peculiar phenotypes that have arisen. In plants, they have engendered horticultural interest because they have produced visually unusual phenotypes, and in agriculture because they have aided production of commercially desirable fruits and vegetables. However, despite occasional observation of their apparent role in insect resistance, no sustained effort has been undertaken to exploit this possibility. This lacuna may be partly due to the failure of the genetic mosaic theory to explain much of the observed intra- and inter-individual variation in insect damage and thus to explain in general how long-lived plants survive evolutionarily in the face of persistent attack by short-generation phytophagous insects.¹⁷ Nevertheless, several examples of resistance to insects generated by such mutations in trees suggest that a more concerted effort to seek out similar cases would be worthwhile, and that exploration is warranted as a means of exploiting resistance that arises by somatic mutations in individuals for more broad-scale defense of populations.

Strikingly, the example of multiple somatic mutations in hydrilla inducing resistance to a formerly highly effective and widely used herbicide has many unexplained features, perhaps most generally,

why have similar cases not originated in other pest plants? One reason is the inherent difficulty in detecting point mutations that do not code for visible phenotypes. A possible explanation for why the hydrilla mutations were detected is the unusual ability of hydrilla to reproduce and establish by plant segments, so that a segment with a novel somatic mutation need not die when a herbicide kills the remainder of an otherwise susceptible plant.³⁴ In any event, these hydrilla mutations combined with modern techniques of genetic manipulation suggest the possibility of producing herbicide-resistant crop plants. Advances in whole genome sequencing and functional genomics³⁵ should increase the detection rate of somatic point mutations.

Observations and research on both insect-resistant trees and herbicide-resistant plants point to a broader, fundamental issue in plant biology and genetics. Despite a long history of observation of somatic mutations and their phenotypic effects, many research gaps remain. The dearth of examples in both areas – insect resistance and herbicide resistance – points to the major question – why so few known cases, and have features of particular plant-herbivore interactions led to these few cases? An examination of single-nucleotide variants from different locations on two terminal branches of a 234-year old oak tree (*Quercus robur*) revealed between 38 and 47 such variants, suggesting a somatic mutation rate between 4.2 and 5.2×10^{-8} substitutions/site/generation.³⁶ Noting values for *Arabidopsis thaliana* between 7.0 and 7.4×10^{-9} substitutions/site/generation, and the much greater distance (>100-fold) between the two oak branches compared to the distance between the physical location of any two mutant *Arabidopsis* cells, the authors found the observed oak mutation rate more than 10 times less than expected. As a possible explanation for this deficit, they cite research³⁷ showing that, in *Arabidopsis* and tomato, the number of divisions in the cell lineage leading from the shoot apical meristem to axillary meristems is greatly reduced relative to the number of divisions leading to differentiated cells. If this mitosis model were to hold for trees, it might partly explain the small number of observed, or at least reported, somatic mutations in trees. It is worth observing, however, that the oak study did report many variants, none of which would have been apparent in the absence of molecular methods. Given the millions of trees on the face of the earth, the paucity of reported phenotypic somatic mutations in trees remains puzzling. The circumstances under which somatic mutations enter gametophytes, and the frequency with which this happens, remains understudied, and research to elucidate these matters would probably shed light on the previous question.

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REFERENCES

- 1 D'Amato F, Role of somatic mutations in the evolution of higher plants. *Caryologia* **50**:11–15 (1997).
- 2 Klekowski EJ Jr, Plant clonality, mutation, diploidy selection and mutational meltdown. *Biol J Linn Soc* **79**:61–67 (2003).
- 3 Whitham TG and Slobodchikoff CN, Evolution by individuals, plant-herbivore interactions, and mosaics of genetic variability: the adaptive significance of somatic mutations in plants. *Oecologia* **49**:287–292 (1981).
- 4 Marcotrigiano M, Herbivory could unlock mutations sequestered in stratified shoot apices of genetic mosaics. *Am J Bot* **87**:355–361 (2000).

- 5 Conway SJ and Drinnan AN, Analysis of surface growth in the conifer shoot apical meristem. *Int J Plant Sci* **178**:273–287 (2017).
- 6 Klekowski EJ Jr, Lowenfeld R and Klekowski EH, Mangrove genetics. IV Postzygotic mutations fixed as periclinal chimera. *Int J Plant Sci* **157**:398–405 (1996).
- 7 Duke SO, Biotechnology: herbicide-resistant crops, in *Encyclopedia of Agriculture and Food Systems*, Vol 2, ed. by Van Alfen N. Elsevier, San Diego, CA, pp 94–116 (2014).
- 8 Krishna H, Alizadeh M, Singh D, Singh U, Chauhan N, Eftekhari M *et al.*, Somaclonal variations and their applications in horticultural crops improvement. *3 Biotech* **6**:1–18 (2016).
- 9 Whitham TG, Host manipulation of parasites: within-plant variation as a defense against rapidly evolving pests, in *Variable Plants and Herbivores in Natural and Managed Systems*, ed. by Denno RF and McClure MS. Academic Press, New York, NY, pp 15–41 (1983).
- 10 Gill DE and Halverson TG, Fitness variation among branches within trees, in *Evolutionary Ecology*, ed. by Shorrocks B. Blackwell Scientific, Oxford, pp 105–116 (1984).
- 11 Gill DE, Individual plants as genetic mosaics: ecological organisms versus evolutionary individuals, in *Plant Ecology*, ed. by Crawley MJ. Blackwell Scientific, Oxford, pp 321–343 (1986).
- 12 Edwards PB, Wanjura WJ, Brown WV and Dearn JM, Mosaic resistance in plants. *Nature* **347**:434 (1990).
- 13 Padovan A, Keszei A, Wallis IR and Foley WJ, Mosaic eucalypt trees suggest genetic control at a point that influences several metabolic pathways. *J Chem Ecol* **38**:914–923 (2012).
- 14 Padovan A, Lanfear R, Keszei A, Foley WJ and Külheim C, Differences in gene expression within a striking phenotypic mosaic *Eucalyptus* tree that varies in susceptibility to herbivory. *BMC Plant Biol* **13**:29 (2013).
- 15 Padovan A, Patel HR, Chuah A, Huttley GA, Krause ST, Degenhardt J *et al.*, Transcriptome sequencing of two phenotypic mosaic *Eucalyptus* trees reveals large scale transcriptome remodeling. *PLoS ONE* **10**:1–20 (2015). <https://doi.org/10.1371/journal.pone.0123226>.
- 16 Lämke JS and Unsicker SB, Phytochemical variation in treetops: causes and consequences for tree-insect herbivore interactions. *Oecologia* **187**:377–388 (2018).
- 17 Herrera CM, *Multiplicity in Unity. Plant Subindividual Variation and Interactions with Animals*. University of Chicago Press, Chicago (2009).
- 18 O'Connell LM and Ritland K, Somatic mutations at microsatellite loci in western redcedar (*Thuja plicata*: Cupressaceae). *J Hered* **95**:172–176 (2004).
- 19 Havill NP, Shiyake S, Galloway AL, Footitt RG, Yu G *et al.*, Ancient and modern colonization of North America by hemlock woolly adelgid, *Adelges tsugae* (Hemiptera: Adelgidae), an invasive insect from East Asia. *Mol Ecol* **25**:2065–2080 (2016).
- 20 Eschtruth AK, Evans RA and Battles JJ, Patterns and predictors of survival in *Tsuga canadensis* populations infested by the exotic pest *Adelges tsugae*: 20 years of monitoring. *Forest Eco Manag* **305**:195–203 (2013).
- 21 Painter RH, Lessons to be learned from past experience in breeding plants for insect resistance, in *Breeding Pest Resistant Trees*, ed. by Gerhold HD, Schreiner EJ, McDermott RE and Winieski JA. Pergamon, Oxford, pp 349–355 (1966).
- 22 Bobiwash K, Schultz S and Schoen D, Somatic deleterious mutation rate in a woody plant: estimation from phenotypic data. *Heredity* **111**:338–344 (2013).
- 23 Barrett SCH, Influences of clonality on plant sexual reproduction. *Proc Natl Acad Sci U S A* **112**:8859–8866 (2015).
- 24 Michel A, Arias RS, Scheffler BE, Duke SO, Netherland M and Dayan FE, Somatic mutation-mediated evolution of herbicide resistance in the nonindigenous invasive plant hydrilla (*Hydrilla verticillata*). *Mol Ecol* **13**:3229–3237 (2004).
- 25 Saltonstall A and Meyerson LA, *Phragmites australis*, from genes to ecosystems. *Biol Inv* **18**:2415–2420 (2016).
- 26 Mariani C, Cabrini R, Danin A, Piffanelli P, Fricano A, Gomasasca S *et al.*, Origin, diffusion and reproduction of the giant reed (*Arundo donax* L.): a promising weedy energy crop. *Ann Appl Biol* **157**:191–202 (2010).
- 27 Bailey JP, Bimova K and Mandak B, Asexual spread versus sexual reproduction and evolution in Japanese Knotweed s.l. sets the stage for the “Battle of the Clones”. *Biol Inv* **11**:1189–1203 (2009).
- 28 Holm A-K, Elameen A, Oliver BW, Brandsæter LO, Fløistad IS and Brurberg MB, Low genetic variation of invasive *Fallopia* spp. in their northernmost European distribution range. *Ecol Evol* **8**:755–764 (2018).
- 29 Bzdęga K, Janiak A, Tarłowska S, Kurowska M, Tokarska-Guzik B and Szarejko I, Unexpected genetic diversity of *Fallopia japonica* from Central Europe revealed after AFLP analysis. *Flora* **207**:636–645 (2012).
- 30 Bzdęga K, Janiak A, Książczyk T, Lewandowska A, Gancarek M *et al.*, A survey of genetic variation and genome evolution within the invasive *Fallopia* complex. *PLoS ONE* **11**:1–23 (2016). <https://doi.org/10.1371/journal.pone.0161854>.
- 31 Puri A, MacDonald GE, Haller WT and Singh M, Phytoene and β -carotene response of fluridone-susceptible and -resistant hydrilla (*Hydrilla verticillata*) biotypes to fluridone. *Weed Sci* **54**:995–999 (2006).
- 32 Arias RS, Netherland MD, Scheffler BE, Puri A and Dayan FE, Molecular evolution of herbicide resistance to phytoene desaturase inhibitors in *Hydrilla verticillata* and its potential use to generate herbicide-resistant crops. *Pest Manag Sci* **61**:258–268 (2005).
- 33 Puri A, Haller WT and Netherland MD, Cross-resistance in fluridone-resistant hydrilla to other bleaching herbicides. *Weed Sci* **57**:482–488 (2009).
- 34 Dayan FE, Owens DK, Tranel PJ, Preston C and Duke SO, Evolution of resistance to phytoene desaturase and protoporphyrinogen oxidase inhibitors-state of knowledge. *Pest Manag Sci* **70**:1358–1366 (2014).
- 35 Alonso JM and Stepanova AN eds, *Plant Functional Genomics*, 2nd edn. Springer, New York (2015).
- 36 Schmid-Siegert E, Sarkar N, Iseli C, Calderon S, Gouhier-Darimont C, Chrast J *et al.*, Low number of fixed somatic mutations in a long-lived oak tree. *Nat Plants* **3**:926–929 (2017).
- 37 Burian A, Barbier de Reuille P and Kuhlemeier, patterns of stem cell divisions contribute to plant longevity. *Curr Biol* **26**:1385–1394 (2016).